ECON 0100 | Demo C

Demos are similar to Checkpoints, often taken directly from past semesters. Work through the problems and check your work against mine. Practice answering clearly and completely. You are encouraged to work in small groups. Find a study room, grab some classmates, and work together on a whiteboard.

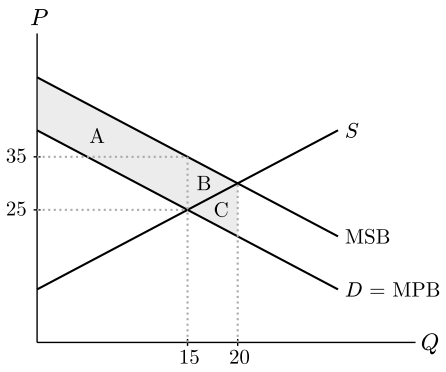
Patronus Lessons

Every winter, dementors drift through Hogsmeade, and the surest defense against them is the Patronus Charm. Private tutors sell lessons in the charm. Each witch or wizard who learns it also protects their neighbors from dementors, an external benefit of 10 coins per lesson. Prices are in coins and quantity is in lessons. The Demand (Marginal Private Benefit) and Supply (Marginal Cost) equations for lessons are the following:

$$D : P_b = 40 - Q_d \qquad S : P_s = 10 + Q_s$$

C1.2 | Patronus Lessons

The graph shows the market for lessons: marginal private benefit (MPB), marginal social benefit (MSB), and supply (S). The lettered regions are for the questions.



MSB lies _____ coins above MPB at every quantity
Market Quantity ($\hat{Q}$): _____
Efficient Quantity (Q^*): _____
Deadweight Loss (*letter*): _____
Deadweight Loss (*area*): _____
MSB of the Last Lesson Traded: _____
MC of the Last Lesson Traded: _____
The Market Trades: ☐ too many ☐ too few lessons

C2.2 | A Subsidy for Lessons

To encourage lessons, the Ministry pays tutors a subsidy of 8 coins per lesson. For this question, ignore the external benefit: demand is the marginal benefit (MB) of a lesson.

Buyer's Price (P_b): _____
Seller's Price (P_s): _____
Subsidy Quantity (Q^{Sub}): _____
Deadweight Loss (DWL): _____
Last Lesson Traded, MC vs. MB: ☐ above ☐ below

Flummoxed by the Floo Powder Fiasco

Floo Powder is a magical commodity used by witches and wizards to transport themselves through fireplaces to different locations. While it's often convenient, its use is associated with negative environmental effects, including air pollution and the depletion of magical resources used to produce the powder. The Ministry's Bureau of Economic Analysis has estimated the externality to be 12 coins per tin with the following supply and demand equations:

$$D : P_b = 120 - 2Q_d \qquad S : P_s = Q_s$$

C1.1 | The Fiasco

Start by using a graph and algebra to describe the market failure.



Graph demand, supply, and MSC, and shade the deadweight loss.
Marginal Social Cost (MSC): _____
Market Quantity ($\hat{Q}$): _____
Efficient Quantity (Q^*): _____
Deadweight Loss (DWL): _____
The Market Trades: ☐ too much ☐ too little

C2.1 | A Tax for Revenue

Needing revenue, and not yet knowing the size of the externality, the Ministry imposes a tax of 6 coins per tin on Floo Powder. For this question, ignore the external cost.



Buyer's Price (P_b): _____
Seller's Price (P_s): _____
Post-Tax Quantity (Q^{Tax}): _____
Tax Incidence on Buyers: _____
Tax Incidence on Sellers: _____
Deadweight Loss (DWL): _____
Who Bears More of the Tax? ☐ buyers ☐ sellers ☐ equal

C3.1 | Designing the Policy

The Ministry of Magic has hired you to design a policy that corrects the market failure caused by the negative externality of Floo Powder use. Propose the policy you recommend and how it would impact the market for Floo Powder and its users.

Policy Type: _____
Policy Size: _____
Post-Policy Quantity (Q^τ): _____
Post-Policy Deadweight Loss: _____
Post-Policy Quantity vs. Efficient: ☐ above ☐ at ☐ below